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Personalized recommendation system based on product specification values

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Abstract

In this paper, we developed a recommendation system which enables bidirectional communication between the user and system using an utility range-based product recommendation algorithm in order to provide more dynamic and personalized recommendations. The system is based on an interactive procedure for recommending similar ones among the products of the collaborative companies that share the product taxonomy table. The main idea of the proposed procedure is using a multi-attribute decision making (MADM) to find the utility values of products in same product class of the companies. Based on the values, we determine what products are similar. The similar product recommendation system is a Web-based application system running on a PC. The system has a user-friendly graphic user interface to encode easily incomplete value judgments. Using the system, we carry out the experiments for performance evaluation of our procedure. The experimental study shows that the utility range-based approach is a viable solution to the similar product recommendation problems in the viewpoints of correct rate and satisfaction rate.
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Keywords: Personalized recommendation; Similarity measure; Collaborative commerce; Incomplete information

1. Introduction

In recent trend of digital economy, enhanced interaction between the organizations and its customers is recognized as the key characteristic of the marketing environment, and firms are facing the challenges of providing differentiated products and services to each customer based on the information they collect from the customers. The terminology that best describes this phenomenon is personalization. According to Allen, Kania, and Yaeckel (1998), firms can maintain the close relationship with the customers and raise customer loyalty by offering differentiated service based on personalization function. And, personalization at the content providing site can be used as a strategy of helping users save time and costs of searching the contents, and as a strategy to maximize revenue by increasing the level of user satisfaction about the site. Personalization is now even used in the area of product recommendation, and a number of researches are being conducted on personalized product recommendation system. However, there has not been any research on recommendation systems based on high involvement product which is characterized as involving low purchasing frequency, high price, and many evaluation criteria and much time in making a purchasing decision (Blackwell, Paul, & James, 2000). Personal computer (PC), automobile, and furniture are examples of high involvement products, and consumers have the tendencies to actively search information about the products, compare and evaluate the alternatives carefully, and consider few specific criteria with heavy emphasis among several available evaluation criteria. Therefore, the customers actively provide more detailed preference information in the searching process of high involvement product.

Furthermore, prior research on personalized product recommendation system lacks the academic rigor on the following issues. First, the recommendation system does not efficiently incorporate the each user's preference. It is not possible to take account of each user's preference although existing recommendation systems suggest products from the people who have similar preferences, using a number of similarity measures for product recommendation. There are not many ways to reflect the preference, for example, when buying a computer, brand is an important factor in purchasing decision

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for some users, but others consider price as more important factor in his or her preference. Second, there is a problem of recommendation method being implemented in a one-sided way, mainly from a system to the user. For example, general purchase process at a store incurs through the communication between the customer and sales clerk. Through this effective communication, transaction takes a place when a customer acquires the information he or she wants and gives the requirements about the product to the sales clerk. But, most systems until now lacked the process of re-handling based on real time feedback, they merely suggested recommended product in a one-sided way.

Therefore, in our research, we developed a recommendation system which enables bidirectional communication between the user and system using an utility range-based product recommendation algorithm in order to provide more dynamic and personalized recommendations. Consequently, we suggested the business process for implementing the business model of electronic commerce that is based on collaboration among the firms and conducted an empirical experiment to test the system and relevant algorithm.

2. Prior research on similarity measure for recommendations

Recommendation system is defined as the system which recommends an appropriate product or service after learning the customers' tendency and desire. Today, a number of researchers are conducting a rigorous study on recommendation system, and the most important factor in recommendation system is an ability to correctly filter and analyze the customer preference and behavior, thereby recommending the best product which the customer wants based on accurate estimation approach (Cho, Kim, & Kim, 2002; Sarwar, Karypis, Konstan, & Reidl, 2000, 2001). This leads to the research on the personalized recommendation method, and two of the well-known representative methods are content based filtering and collaborative filtering (Goldberg, Nichols, Oki, & Terry, 1992; Mooney & Roy, 2000).

Montaner, Lopez, and Rosa (2003) suggest advanced classification of Internet-based intelligent recommendation system, and they distinguish the recommendation methods into content-based filtering, collaborative filtering, and hybrid of which previous two methods are combined. And, cosine similarity, naive bayesian (NB) classifier, Pearson *r* correlation, etc. are being used as key similarity measure for recommendation. Table 1 below reorganized the recommendation systems in terms of similarity measure, recommendation method, and recommendation domain.

But, the key similarity measures, which are used in the recommendation system, such as cosine similarity, Pearson *r* correlation, NB classifier, Euclidean distance, etc. have the following weaknesses. In case of cosine similarity measure, it is possible to apply the measure when records for clustering are separated out from the origin and the clusters are away from (Schalkoff, 1992). In case of Pearson *r* correlation, problem of scalability occurs where algorithm processing speed slows down when there is a high number of users and the item types

Table 1
Classification of recommendation system

Method	System	Domain	Similarity measure
Content based filtering	Amalthaea	Web recommender	Cosine similarity
	Let's browse	Web recommender	Cosine similarity
	Letizia	Web recommender	Cosine similarity
	Syskill and Webert	Web recommender	NB classifier
			Nearest neighbor
			Cosine similarity
	Webmate	Web recommender	Cosine similarity
	LaboUr	Document recommender	NB classifier
			Nearest neighbor
	MovieLens	Movie recommender	Cosine similarity
	News dude	Netnews recommender	Nearest neighbor
			Cosine similarity
			NB classifier
	NewT	Netnews filtering	Cosine similarity
	Re:agent	E-mail filtering	Nearest neighbor
Collaborative filtering	Bellcore video recon	Movie recommender	Pearson *r* correlation
	GroupLens	News recommender	Pearson *r* correlation
	LaboUr	Document recommender	Pearson *r* correlation
	MovieLens	Movie recommender	Cosine similarity
	Ringo/firefly	Music recommender	Pearson *r* correlation
	Smart radio	Music lists recommender	Pearson *r* correlation
Hybrid	Fab	Web recommender	Cosine similarity
	Personal WebWatcher	Web recommender	NB classifier
	WebWatcher	Web recommender	Cosine similarity
	Anatagonomy	News recommender	Cosine similarity
	Krakatoa chronicle	News recommender	Cosine similarity
	News weeder	News recommender	Cosine similarity

due to the scarcity problem especially in case of data with a number of item types (Billsus & Pazzani, 1998; Sarwar et al., 2000, 2001). In case of NB classifier, it can be applied when the attributes values are categorical data (Billsus & Pazzani, 1998). And, there is a problem of not considering the weights of the attributes and a limitation exists with the assumption of thinking that attributes are mutually independent of each other for a record. With respect to Euclidean distance each data should be matched by point but it is hard to match the data by point using the categorical data (Schafer, Konstan, & Riedl, 2001; Shyu, Haruechaiyasak, & Chen, 2003; Yang & Pedersen, 1997). And, it is hard to apply to the non-numerical value data and it results in decrease of processing speed since more calculation procedures are needed as the amount of data increases.

There are merely personalized similarity measures that imply weights of attributes differently for each individual. In this research, we utilize an utility range-based product recommendation algorithm which is based on the concept of additive utility where customer preference regarding a number of product specification values can be expressed in numerical values (Choi & Cho, 2004). An utility range-based product recommendation algorithm as a method which is applicable in cases of small amounts of data can be utilized in problems with different weights by each product specification. And, such weights of the product specifications imply the user preferences. For example, when purchasing the same computer, if the customers consider CPU type, brand, price, etc. some customers consider CPU type as the most important factor influencing his or her purchase decision but others might consider brand as most important factor. As such, customers in general have preference information which is not the same regarding the product specification (spec) and it is difficult to measure the weights of product spec as a specific numerical value. Here, the basic concept of algorithm suggests that it is more efficient to provide incomplete information of which customers express weights of product specifications in terms of the mutual relationships among the weights, and refer to the study done by Choi, Kim, and Kim (1999) for more detailed description on incomplete information.

3. Similar product recommendation problems and process

3.1. Applicable business model

An utility range-based product recommendation algorithm has a number of application areas since it enables more personalized recommendation by applying the different weights of spec by each customer. Some of the potential application areas are as follows. First, firms can recommend the products with specification values which the customer desires when the firms want to recommend a product with a particular specification value. Second, mutually complementary product recommendation is possible between the collaborative companies for the products that are in the identically defined product class by sharing the product map between the collaborative companies. Third, it can be applied to the intermediary agent system for negotiation in order to maximize the mutual benefits of two agents participating in the negotiation with a consideration of multiple product features. Forth, recommendation of most similar alternative product is possible when the product which the customer wants to purchase from a particular company is not in stock. The business model, which we discuss in this research, is the second application area. The detailed procedure of the collaborative commerce model is suggested by Choi and Cho (2004).

3.2. Problem definition for similar product recommendation

An utility range-based product recommendation algorithm suggests a way to classify the products in the class of Stock Keeping Units (SKUs), which can be defined by same set of product specifications, according to the user preference. Using this algorithm, it can be distinguished into class with same set of specifications. And, based on the specification value of each SKUs, detailed classification can be conducted. Here, specification value converts to user's utility value and each product's utility value can be calculated by computing the sum of weights based on weights of each specification. Similarly, it becomes possible to apply the different preference of each user by taking an account of different weights of specification for each user.

The products that belong in a same class can be evaluated based on a set of product specifications at the level of SKU for the products currently owned by a firm, and each product can have a various specification values. We define the terminologies as such in order to calculate the utility values of the products in the same class.

$I = \{i\}_{i=1,N}$ a set of N products at the same class
$J = \{j\}_{j=1,M}$ a set of M product specifications at the same class
w_j importance of jth specification
x_{ij} jth specification value of ith product

Multi-criteria decision making (MCDM) can be effectively utilized in evaluation of the products with limited attributes since, it is a method of evaluating the alternative decisions with finite number of attributes. A well known method is using it as a weighted additive form in terms of utility function in order to evaluate the product, $x_i = \{x_{i1}, x_{i2}, \ldots, x_{iM}\}$ which has M number of attribute values.

$$v(x_i) = \sum_{j=1}^{M} w_j v_{ij}(x_{ij}) \tag{1}$$

Here, v_{ij} is the marginal value function of specification j such that v_{ij}: $x_{ij} \rightarrow [0, 1]$. After all, the objectives of an utility range-based product recommendation algorithm are to calculate the utility range of the products, which the collaborative company owns and belongs in the same class as the products selected by the customers and suggest similar products to the customer which has the closest utility value as the product, x_s, which he or she selected. If both the decision parameters, w_j and $v_{ij}(\cdot)$ are given as the accurate numerical values by a

decision maker in order to calculate the utility values of the products then utility functions values, x_s and x_i can be easily calculated using the Formula (1). But, it is difficult to calculate the accurate values of the weights in the real problems. Instead, it is easy for a decision maker to provide the information about the weights in format of incomplete information in terms of relationships among the weights and ranges of weight values (Choi & Cho, 2004; Choi, Kim, & Kim, 1999).

3.3. Similarity measure

This section suggest, the similarity measure that show the level of similarity between the two products x_s and x_i using the utility value based on the individual's personal preference. As discussed in the previous section, it is difficult to calculate the accurate utility value of a product, this research tries to calculate the utility ranges of product and assumes the ranges for product x_s and product x_i as $[v_s(\min), v_s(\max)]$ and $[v_i(\min), v_i(\max)]$ respectively. Similarity measure between the two products can be defined by the ratio of overlapping range of two products' utility values and by the level of closeness between the average utility values. The relationship based on utility values between the two products can be explained in following six cases. There is no overlapping area for Case 1 and Case 2. The overlapping range for Case 3 falls in the range for the product x_s. And for Case 4 and Case 5, a part of the ranges for both products overlaps. Lastly, overlapping range for Case 6 covers the overall range of product x_s.

The ratio of range overlap between the two products x_s and x_i is calculated as the following. If it happens that $v_i(\max) \leq v_s(\min)$ or $v_i(\min) \geq v_s(\max)$, then there exists no overlapping range for Case 1 and Case 2, but other Cases, overlapping range can be calculated by [maximum between $v_i(\min)$ and $v_s(\min)$, minimum between $v_i(\max)$ and $v_s(\max)$]. The ratio of overlapping range between the two products is calculated by the following formula, which is the ratio of the width of overlapping range to that of sum of ranges of two products.

$$\mathrm{or}_{\mathrm{si}} = \frac{2\{\min[v_i(\max), v_s(\max)] - \max[v_i(\min), v_s(\min)]\}}{[v_s(\max) - v_s(\min)] + [v_i(\max) - v_i(\min)]}$$

for $i = 1, \ldots, N$

Here, the value of $\mathrm{or}_{\mathrm{si}}$ falls in the range between 0 and 1, and higher value means better the similarity. Then, in Fig. 1, the products in Cases 3, 4, 5, and 6 show more similarity than the products in Case 1 or Case 2 with respect to product x_s.

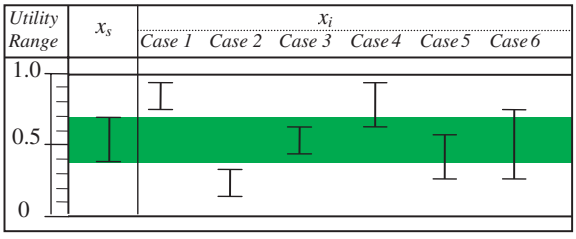


Fig. 1. The relationship based on utility value ranges between the two products.

The level of closeness to the average value used as a second item of similarity measure is calculated based on the difference of average values. But, since it is difficult to calculate the accurate average values for the utility values of the products, we use the median value instead. The level of closeness to the average value between the two products can be calculated as the following.

$$\mathrm{pr}_{\mathrm{si}} = 1 - \left| \frac{[v_s(\min) + v_s(\max)]}{2} - \frac{[v_i(\min) - v_i(\max)]}{2} \right|$$

for $i = 1, \ldots, N$.

Here, the value of $\mathrm{pr}_{\mathrm{si}}$ falls in the range between 0 and 1, higher value means better the similarity. Finally, level of similarity between the two products can be calculated as the following.

$$\mathrm{sim}_{\mathrm{si}} = \alpha\, \mathrm{or}_{\mathrm{si}} + (1 - \alpha)\mathrm{pr}_{\mathrm{si}} \quad (2)$$

In this case, we can assign value to α in the range of 0 and 1 and it can be defined as an importance constant of overlapping ranges.

3.4. Similar product recommendation procedure

This procedure is applied when a customer searches for a similar product, which is in the identical category, among the products at the collaborative site for the product he or she selected at the purchasing location. The process is based on the business model where product class and spec information are shared between the shopping malls of two different countries that are under collaborative relationship. Similar product searching process can be described as seen in Fig. 2.

First, the purchasing customer enters the order information of the product he or she wants at the domestic shopping mall. Then, when the customer selects the delivery location, order information is processed through the normal ordering process if the product recipient is the purchaser or other domestic resident. But, if the recipient is a resident of another country, search for the same product having the same product specification values is conducted using the product list of the site at the delivery location, which is in collaborative relationship with the site where the product search is handled. At the time, if the same product exists, customer purchases the product at the shopping mall and makes a specific delivery request. But, if there is no same product, the process of similar product recommendation is followed.

The process of similar product recommendation involves having a customer enter the relationship among the weights by product spec based on the five types of expressing the incomplete information and then it requires the assigning of the number of similar products based on recommendation. Up to this point is the stage of receiving the system input information from the customers, and we apply an utility range-based product recommendation algorithm based on those information. The detailed processing order of an utility range-based product recommendation algorithm consists of following five steps.

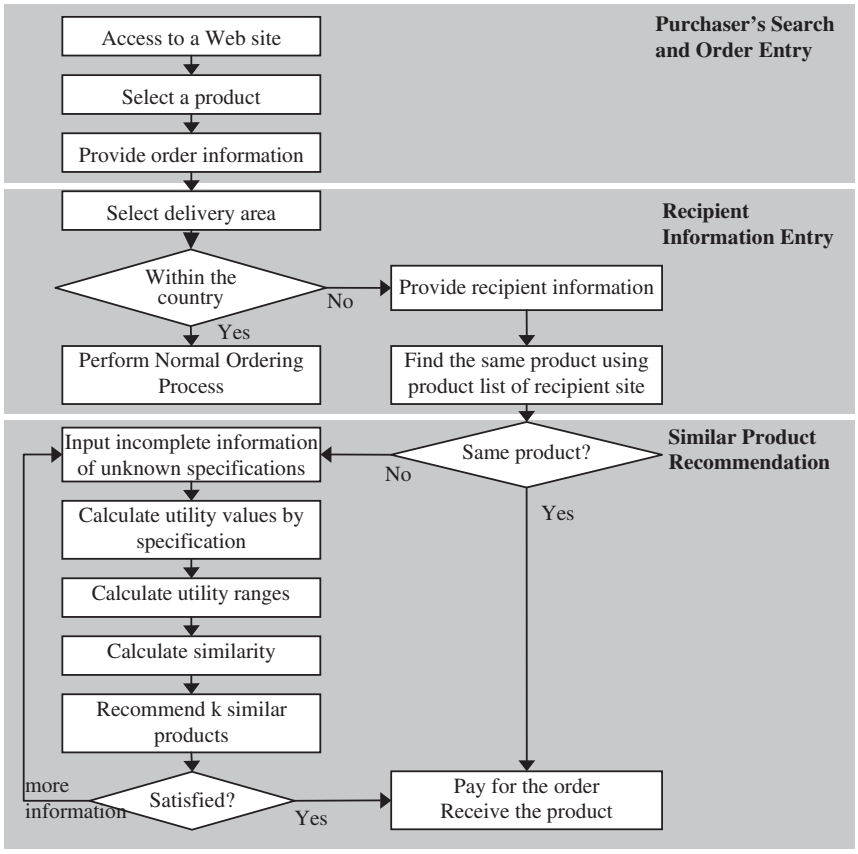


Fig. 2. Similar product recommendation procedure.

Step 1: *Calculation of utility value by specification*—Add the specification values of the products of which one is selected by the customer and others are kept at the receiving location and calculate the utility value with respect to each specification based on the added specification value. At this time, utility value v_{ij} for ith product and jth specification can be calculated based on formula (3) and it is expressed as value in the range of [0, 1].

For specification j with better for larger,

$$v_{ij} = \frac{x_{ij} - \min_i x_{ij}}{\max_i x_{ij} - \min_i x_{ij}}$$

and for specification j with better for smaller

$$v_{ij} = \frac{x_{ij} - \max_i x_{ij}}{\min_i x_{ij} - \max_i x_{ij}} \tag{3}$$

Step 2: *Calculation of expected utility value by product*—Calculate the range of utility values for the products in the same class as the customer selected product at the product list of the receiving location. In this case, it is difficult to have accurate value of weight w_j for specification value to be entered by a decision maker. Therefore, it is desired to obtain the weighted additive ranges for utility value in terms of solving the linear programming problem by using the incomplete information between the weights of product specifications which is entered by the decision maker and using it as a set of constraints. And, by using the five types of incomplete information discussed in Section 3.2 and collecting the incomplete information, which is entered as relative weight information for product specification, we define the set of relationships among the weights as Φ_W. Using the information obtained, utility value range, $[v_i(\min), v_i(\max)]$ for ith product is calculated using the following formula.

$$v_i(\min) = \min \sum_j w_j v_{ij} \text{subject to } \Phi_W \tag{4}$$

and

$$v_i(\max) = \max \sum_j w_j v_{ij} \text{subject to } \Phi_W$$

Step 3: *Calculation of similarity measure values between two products*—Calculate the similarity measure values based on utility value ranges of each product from recipient location and customer selected product from Step 2 using the formula (2).

Step 4: *Suggestion of k number of similar products in order of similarity measure values*—Define the product as being highly similar product when sim_{si} value for each product at the receiving location is high, and suggest k numbers of similar products in the descending order of sim_{si} values. In this case, customer selects the values of α and k.

So far, we have examined an utility range-based product recommendation algorithm based on concept of additive utility function in uncertain situation and similar product searching

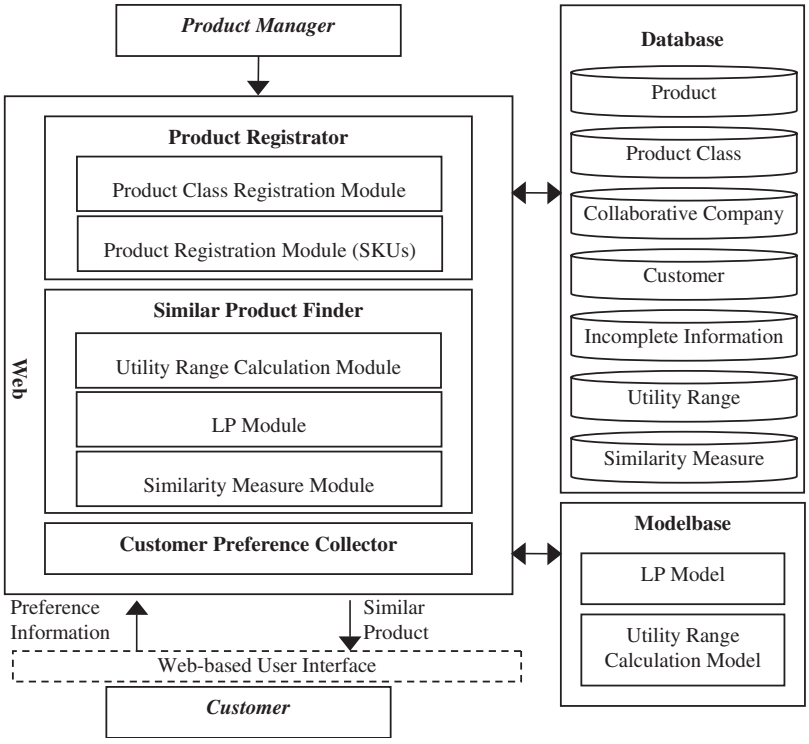


Fig. 3. System architecture.

process. And, we introduce details of system implementation in the next chapter.

4. System implementation

The system largely consists of Product Registrator, Customer Preference Collector, and Similar Product Finder, and further they are made up of several sub-modules. Fig. 3 shows the system architecture of the system in discussion. This system is based on web-based user interface, and the processes in product manager and customer sides are integrated with each relevant module.

4.1. Product registrator

Product Registrator consists of Product Class Registration Module and SKU-based Product Registration Module. Product managers handle the tasks relating to product registration and they define the new product class and decides the specification of each product class. Product manager must first define the product class of a given product before registering the new product unit. If there is predefined product class, the new product unit should be registered under the appropriate class. But if there is not a predefined product class, a new product class should be created according to the following process.

First, define the ID and name of the product class and define the spec of the product class. Similarly, refer to 'quantitative specs' as specs that can be measured such as price, etc. and 'qualitative specs' as specs that are of non-numerical values of specification such as manufacturer name. Next, assign B.F.L.(Better For Larger; for example, computer memory) and B.F.S.(Better For Smaller; for example, price) for each spec. Lastly, the process of assigning the category value for calculating the utility value of qualitative spec is examined. Fig. 4 shows a notebook computer class defined by six specifications.

After product class and detailed spec are defined, new unit products begin to be regularly registered. Then, product registration is divided into product registration of those with known specification value and that of not knowing partial and entire specification value. First, registering the product, of which its specification value is already known, involves entering the numerical value when the specification value under consideration is quantitative specification value. But, if it is qualitative specification value then it selects the corresponding category value. Lastly, specification value is expressed by setting the relationship information among other products as five types of incomplete information when accurate specification value of a product as seen in Fig. 5 is not known. The relationships are used as the constraint when calculating the maximum and minimum values of specification value through LP Module.

The relative utility value of each product, which has accurate specification value, is computed using Formula (2) when registration of all the unit products is completed.

4.2. Customer preference collector

In order to recommend a similar product, preference information about the importance rate for each specification should be entered by the customer. But, as mentioned in

Defined Product Class Spec

No.	Spec Name	Category Value	Determine B.F.L. or B.F.S.	Delete
1	Price	Define	B.F.S.	Delete
2	Brand	Defined	B.F.L.	Delete
3	RAM	Defined	B.F.L.	Delete
4	HDD	Defined	B.F.L.	Delete
5	ODD	Defined	B.F.L.	Delete
6	Display	Defined	B.F.L.	Delete

※ B.F.L : Better For Larger
B.F.S : Better For Smaller

Fig. 4. Defined product class spec.

Section 3.2, preference information is usually incomplete and is stated inaccurately. Therefore, we developed and utilized the tool in this system in order to effectively express the incomplete information of specification weights.

The incomplete information is encoded in the input screen of the software through the use of a user-friendly interface as shown in Fig. 6. The capture screen displays the portion of a participant's incomplete value judgment about relations between the specifications. For example, the screen shows that price spec is more important than each spec of brand, RAM, HDD, ODD, and display type. Then, we get the relationship between the specifications as constraints of formula (4) and the value range of *i*th computer as the result.

4.3. Similar product finder

This module help the system go through the process of maximization and minimization through LP Module in order to calculate the range of utility value which is used for measuring product similarity. This system supports the automated formulization of linear programming problem, and this is enabled by LP Module. Solution for the formulized linear

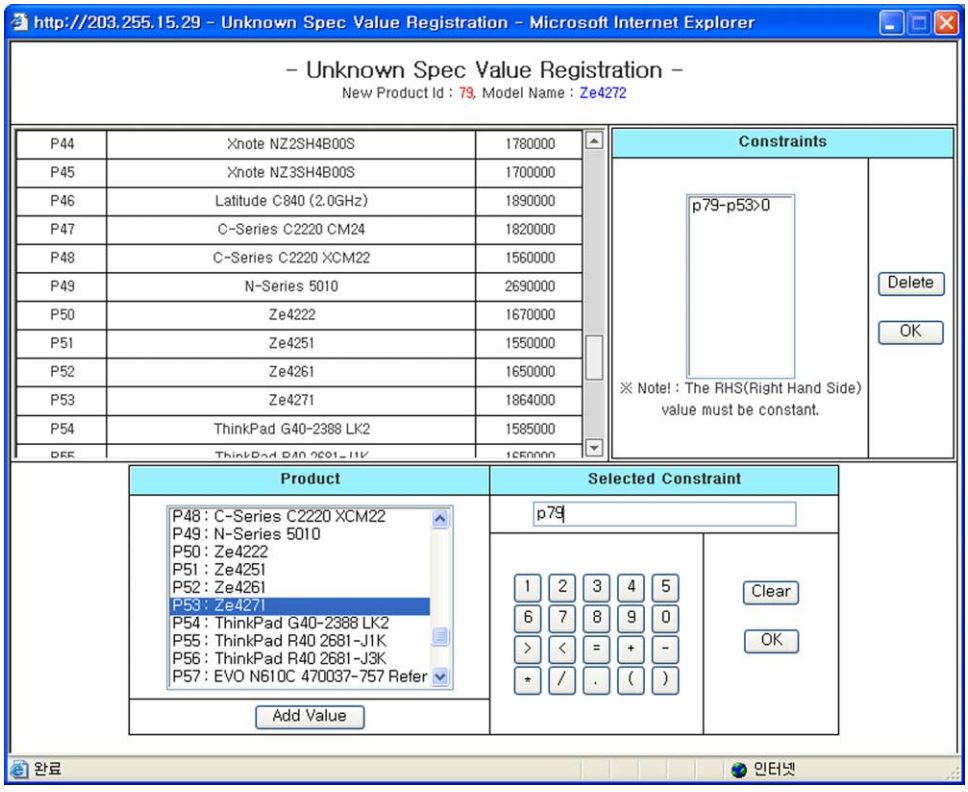


Fig. 5. Unknown spec value product registration.

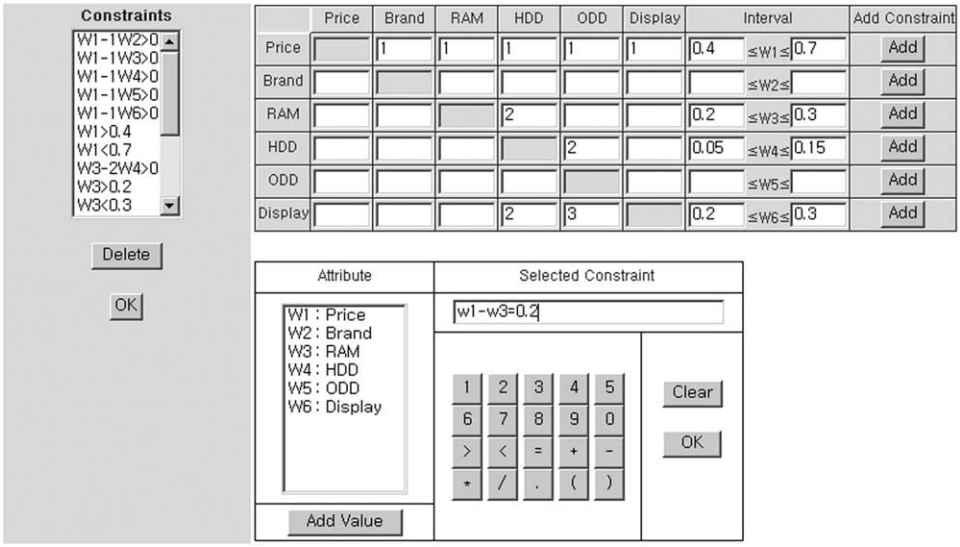


Fig. 6. Screen of incomplete information entry.

programming problem is searched through Lindo API which is an internal application. And, calculated maximum value and minimum value are stored in the database by each product. According to the similar product searching process as shown in Fig. 2, we specify the number of similar products with the product which the customers want and similar product is suggested through internal processing of each module. Fig. 7 represents the screen showing the result of similar product finding.

5. Experimental study

We conducted an empirical experiment to measure the level of user satisfaction and accuracy of an utility range-based product recommendation algorithm which is being discussed in this research.

5.1. Experiment setting

The experiment was conducted with 8 students who did not have prior knowledge about the algorithms used in the system and who had different preferences. And, we used the utility range-base algorithm together with recommendation method of Euclidean distance measure as a comparison method. In terms of evaluation criteria used in the experiment, each user was asked to directly select the choice using the seven point Likert scale (1, Very inaccurate; 4, Average; 7, Very accurate) to indicate the degree of identicalness between the user selected product and recommended similar product. And, using

[Similar Product Finding - Result]

You selected the following product at the L mall:

Product ID	Model Name	Price	Brand	RAM	HDD	ODD	Display
54	ThinkPad G40-2388 LK2	1585000	LG-IBM(2)	DDR512(5)	60GB 이상(5)	COMBO(5)	14inch 이상 ~ 16inch 미만(3)

We have found 5 similar products at the B mall respectively by Utility Range-based Approach.

● Utility Range-based Approach

Product ID	Model Name	Price	Brand	RAM	HDD	ODD	Display
13	PCG-GRX770	1990000	Sony(5)	DDR512(5)	30GB 이상 ~ 60GB 미만(3)	COMBO(5)	16inch 이상(5)
4	EVO N1000C 470051-341	1409000	Compaq(3)	DDR256(4)	30GB 이상 ~ 60GB 미만(3)	COMBO(5)	14inch 이상 ~ 16inch 미만(3)
6	EVO N1020V 470059-789	1630000	Compaq(3)	DDR512(5)	30GB 이상 ~ 60GB 미만(3)	COMBO(5)	14inch 이상 ~ 16inch 미만(3)
23	드림북 N7600.4	1435000	TG(2)	DDR256(4)	30GB 이상 ~ 60GB 미만(3)	COMBO(5)	14inch 이상 ~ 16inch 미만(3)
24	드림북 N7620.4	1435000	TG(2)	DDR256(4)	30GB 이상 ~ 60GB 미만(3)	COMBO(5)	14inch 이상 ~ 16inch 미만(3)

Fig. 7. Result of similar product finding.

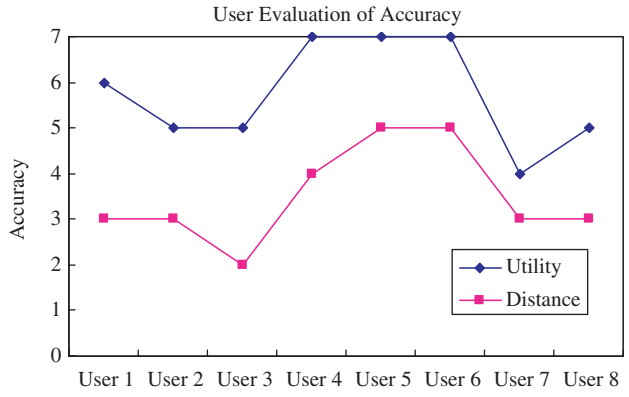


Fig. 8. Result of accuracy evaluation.

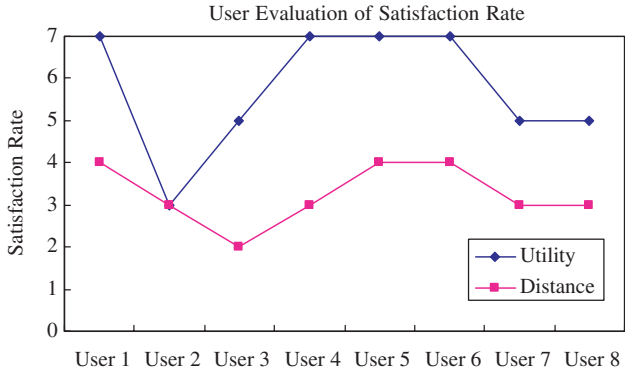


Fig. 9. Result of satisfaction evaluation.

the seven point Likert scale (1, Very dissatisfied; 4, Average; 7, Very satisfied), the user was also asked to indicate the level of satisfaction in the input and output process of using the system while executing the similar product recommendation process. The product class used in the experiment is notebook computer and the relevant product spec considered are limited to price, brand, memory, hard disk capacity, display size, and CD-ROM drive type. For the ease conducting the experiment, registering the product and defining the category values regarding the product spec are previously implemented by system manager.

In the first part, problems and relevant issues were explained to the experiment participants, and the actual experiment was conducted in the following order regarding the similar product recommendation as shown in Fig. 2. Here, we identified the number of similar products which are to be recommended as five. First, an experiment participant enters the personal preference information for each product spec using an appropriate input entry format as shown in Fig. 5. And, experiment indicates the accuracy and satisfaction of the suggested similar products when the system using the entered personal preference information suggests five similar products by each internal module and an utility range-based product recommendation algorithm. Next, we also proceeded in the comparable manner to indicate the accuracy and satisfaction of the suggested similar products by Euclidean distance measure.

5.2. Results of experiment

According to the result of accuracy evaluation and satisfaction evaluation as shown in Figs. 8 and 9 relatively, we were able to verify that the recommendation results of an utility range-based product recommendation algorithm, which is used in this system, were higher in accuracy and satisfaction than the results of Euclidean distance measure method.

We conducted *t*-test of a paired sample in order to statistically verify the effectiveness of the two recommendation methods. As shown in Tables 2 and 3, both the accuracy and satisfaction results were statistically valid within significance level of 0.05 in terms of average differences between two methods. The results came out in such that the method was designed to incorporate the individual preference in the interviews of experiment participants.

6. Conclusion and future research directions

In this research, we developed a system based on an utility range-based product recommendation algorithm and conducted

Table 2
Statistical verification on accuracy

Accuracy	Number of users	Mean	Std. dev.	Std. err.	*T*-value	*P*-value
Utility range-based recommendation	8	5.750	1.165	0.412		
Euclidean distance-based recommendation	8	3.500	1.069	0.378	9.00	0.000
Paired difference	8	2.250	0.707	0.250		

Table 3
Statistical verification on user satisfaction rate

Satisfaction rate	Number of users	Mean	Std. dev.	Std. err.	*T*-value	*P*-value
Utility range-based recommendation	8	5.750	1.488	0.526		
Euclidean distance-based recommendation	8	3.250	0.707	0.250	5.92	0.00
Paired difference	8	2.500	1.195	0.423		

a experimental study to examine the functions of the developed system. The system characteristics can be explained from the perspectives of both the firms that are selling the product and customers who are the users of the system. From the firm perspective, it is possible to register and search the similar products even for the product where its precise specification value is not known. And, mutually complimentary similar product recommendation is possible by composing the product map which is share among the collaborating firms based on this system. From the customer perspective, different preferences by each individual can be fully considered by entering the differentiated preference by each spec and more accurate recommendation by entering additional preference information based on the recommended result can be obtained. Further, the key characteristic is its ability to efficiently enter the personal preference information which is hard to accurately express. Lastly, the overall process of product registration and product recommendation also improved the convenience of both the product manager and customers through the utilization of web-based graphic user interface and automated module.

Further, we achieved more improved accuracy and satisfaction level than previous recommendation method through user experiment using the system. Future research interest can be described as such that system development is needed where specification value searching is limited in order to recommend a product that satisfies an unique specification value, apart from current function of selecting a product and searching its similar product. In order for the system development to be successful, the test should be conducted in an organizational setting through practical application.

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